The 64 Meta-State System: Empirical MRI Validation of TRPI Function Pairings

July 12, 2025

Abstract

This study provides the first empirical validation of the 64 Meta-State System within the Trait Response Personality Indicator (TRPI) framework, leveraging neuroimaging data from the Human Connectome Project. Participants were assigned to meta-states using a blended region-wise similarity metric over prefrontal cortex profiles. Five-fold cross-validation achieved a label-match accuracy of 0.69 ($p = 5.21 \times 10^{-202}$, Cohen's $d = 1.91$), and an adjusted Rand index of 0.59 ($p = 0.001$, Cohen's $d = 78.11$), demonstrating robust and highly significant brain-based discrimination between meta-states. The mean within-group (intra-type) masked similarity was 0.67, indicating substantial anatomical consistency among individuals assigned to the same meta-state. The mean individual trait concordance between brain-derived and self-reported Big Five profiles was 0.57, with a median of 0.65 and a range from -0.23 to 0.80 across states, demonstrating robust but variable alignment at the subject level. These results offer compelling neurobiological evidence for the TRPI model, mapping cognitive meta-states onto reproducible patterns of brain organization. Together, these findings establish TRPI as an empirically validated and theoretically integrated personality framework.

Introduction

The Trait Response Personality Indicator (TRPI) has recently introduced a comprehensive neurocognitive framework proposing the existence of 64 distinct meta-states, each characterized by specific pairings of cognitive functions within the Jungian typological paradigm [1].
According to this framework, personality dynamically emerges from a complex interplay between eight Ego (Perceiving-led) and eight Superego (Judging-led) function pairings.
Each of these pairings adheres strictly to three fundamental compositional rules: (i) each pairing combines precisely one introverted function with one extraverted function; [8] (ii) each pairing integrates one perceiving function with one judging function; [8] and (iii) each pairing is localized entirely within a single cerebral hemisphere.[11]

For instance, the ENTP type is posited to primarily operate from the NeTi–SiFe meta-state, whereas the ESFP type is hypothesized to function predominantly through the SeFi–NiTe meta-state. The matrix created by crossing these eight Ego options with the corresponding eight Superego options yields a robust, neurocognitively plausible system comprising exactly 64 discrete personality meta-states.

This system serves not only as a theoretical framework but as a detailed map of hypothesized lateralized prefrontal cortex (PFC) activity patterns underpinning each cognitive configuration.
Given these detailed predictions, the present study aimed to empirically test the neuroanatomical validity of this 64-state TRPI system. Specifically, we sought to determine whether the theoretically derived meta-states corresponded reliably to measurable neuroanatomical patterns in the PFC, using structural neuroimaging data provided by the Human Connectome Project (HCP) [3]. To achieve this, participants were systematically matched to the best-fitting meta-state based upon individual patterns of deviation in PFC regional brain volumes from the population average. Additionally, we examined the stability of these assignments through rigorous cross-validation and evaluated their correspondence with validated personality measures such as the Big Five personality traits.
The central hypothesis driving this investigation was that if the 64-state TRPI framework accurately captures the fundamental structure of personality, then distinct neuroanatomical signatures within the PFC should emerge clearly and reproducibly across multiple independent validation samples. Moreover, we anticipated meaningful correlations between the identified neuroanatomical states and independently measured psychological traits, thereby providing convergent validation of the underlying TRPI theory.

Through this empirical validation, we seek to bridge classical Jungian typological theory with modern neuroscience, contributing to an integrated model of personality that is both theoretically rigorous and empirically supported.

Methods

Participants and Data Acquisition

We utilized publicly available structural neuroimaging data from the Human Connectome Project (HCP) S1200 dataset [3], consisting of high-resolution magnetic resonance imaging (MRI) scans of over 1,100 neurologically healthy adult subjects aged between 22 and 35 years. The dataset includes comprehensive measures of gray matter volume and cortical thickness across anatomically defined brain regions, providing a robust foundation for examining structural differences associated with personality constructs.

Preprocessing of Neuroanatomical Data

Regional gray matter volume values were extracted from preprocessed Freesurfer-derived segmentations provided by the HCP. We specifically focused on eight bilateral prefrontal cortex (PFC) bundles, namely: dorsolateral (dlPFC), ventrolateral (vlPFC), dorsomedial (dmPFC), and ventromedial (vmPFC) regions. These anatomical bundles were operationally defined using standardized Freesurfer parcellation schemes.

To account for variability due to brain size, each region's volume was normalized by dividing it by the intracranial volume (ICV). Subsequently, these normalized regional values were standardized into z-scores, calculated relative to the mean and standard deviation of the entire study population. This procedure yielded a subject-specific neuroanatomical deviation vector for each participant, capturing individualized variations in PFC anatomy.

Meta-State Assignment Procedure

Participants were systematically assigned to one of the 64 TRPI meta-states by comparing their individual PFC neuroanatomical deviation profiles to empirically defined meta-state centroids. Each centroid represented the mean profile of all subjects assigned to a given meta-state during model training.

To quantify similarity between individual profiles and meta-state centroids, we adopted a blended similarity metric comprising two distinct components:

1. **Pattern Similarity (Pearson correlation coefficient):** Assessed the directional alignment between an individual's anatomical profile and the meta-state centroid. Higher correlations indicated greater alignment of regional anatomical patterns.

2. **Magnitude Similarity (Scaled Euclidean distance):** Measured the overall difference in absolute anatomical deviation between each subject and centroid, with smaller distances indicating closer absolute anatomical proximity.

To ensure neither overall shape (pattern similarity) nor absolute magnitude (distance) of PFC deviation vectors dominated the assignment process, we adopted a blended similarity metric with equal weighting (0.5 Pearson correlation, 0.5 scaled Euclidean proximity). Sensitivity analyses confirmed that strongly favoring one component led to assignment drift and reduced overall model stability, validating the necessity of balanced weighting in this context.

Cross-Validation and Model Evaluation

To rigorously test the generalizability and guard against any circularity or data leakage, we implemented a strict five-fold cross-validation strategy. All normalization, standardization, and meta-state centroid computations were derived exclusively from the training partitions, with no access to the test set. Each subject in the test set was independently assigned to the best-fitting meta-state based solely on their own prefrontal profile and the centroids computed from the training data, ensuring a fully out-of-sample evaluation.

Model validity was evaluated across three critical dimensions:

- **Model Stability (Accuracy):** Measured as the proportion of individuals consistently assigned to the same meta-state across independent folds, reflecting assignment reliability and preventing overfitting to the training sample.

- **Big Five Trait Concordance:** Calculated as the mean correlation between brain-derived meta-state profiles (composite regional scores representing function pairings, e.g., dlPFC $\rightarrow$ Se/Ne + Ti for Extraversion) and self-reported Big Five trait scores. These were computed *post hoc*, after all assignments, and did not inform or influence the model or assignments, preventing criterion contamination.

- **Neuroanatomical Distinctness (Brain-Fit *p*-value):** Computed using a multivariate permutation test to assess whether meta-state groupings exhibited statistically significant, non-random separation in PFC profiles.

We validated the TRPI meta-state model using a strict out-of-sample cross-validation pipeline:

1. **Data Preprocessing:** For each subject, gray matter volumes from eight bilateral prefrontal cortex (PFC) bundles and two thalamic regions were extracted, normalized by intracranial volume, and standardized to z-scores using means and standard deviations from the training partition only.

2. **Feature Scaling:** Bundle features were min-max scaled to $[0, 1]$ using training set statistics; the same transformation was applied to the test set.

3. **Meta-State Labeling:** Each subject was assigned a meta-state label by mapping their two most dominant PFC bundles to canonical TRPI function pairings, with hemisphere and function-class rules enforced. Perceiver/Judger status was determined by hemisphere-specific z-score comparisons.

4. **Centroid Computation:** For each meta-state, the training set centroid was computed as the mean of its assigned subject vectors.

5. **Test Set Assignment:** Test subjects were matched to the closest training centroid using a blended similarity metric (mean of Pearson correlation and scaled Euclidean distance), restricted by the dominant and supporting regions.

6. **Validation Metrics:** Accuracy and adjusted Rand index were computed on the test set; significance was assessed using permutation tests (1,000 iterations). All normalization, scaling, and centroid calculations were restricted to the training set within each fold.

Group-Level and Individual Big Five Trait Concordance Calculation

To quantify the correspondence between neuroanatomical meta-state assignments and self-reported Big Five traits, we computed trait profile concordance at both the group and individual levels.

For each test subject, a brain-derived Big Five trait profile was constructed using composite PFC region volumes as follows:

- **Extraversion:** Sum of left and right dorsolateral PFC (L_dlPFC + R_dlPFC)
- **Conscientiousness:** Sum of left and right ventrolateral PFC (L_vlPFC + R_vlPFC)
- **Agreeableness:** Sum of left and right dorsomedial PFC (L_dmPFC + R_dmPFC)
- **Neuroticism:** Sum of left and right ventromedial PFC (L_vmPFC + R_vmPFC), divided by total PFC volume for normalization due to disparity in size relative to other regions.
- **Openness:** Defined as the mean of the above four brain-derived trait values

Each individual's actual trait profile was taken from their normalized NEO-FFI self-report scores.

Group-level concordance: For all test-set subjects with valid data, the group-mean profile for both brain-derived and self-report trait vectors was calculated. The similarity between these two mean profiles was quantified using a blended similarity metric: the average of Pearson correlation and scaled Euclidean proximity (i.e., $0.5\times$ Pearson $+0.5\times$ [1/(1 + Euclidean distance)]). This yields a value between 0 and 1, indicating the overall alignment of group-level brain-derived and actual trait profiles. In this study, the group-level profile concordance was high (0.92), indicating close alignment between the average brain-derived and self-reported trait profiles.

Individual-level concordance: For each subject, the similarity between their brain-derived and self-report trait profiles was also computed using the same blended metric. Across all valid subjects, the average individual concordance was 0.57, with a median of 0.65, and a range from -0.23 to 0.80. This demonstrates substantial but variable subject-level alignment, with some exhibiting individual concordance near or below zero, highlighting heterogeneity in the anatomical-personality mapping.

Hierarchical Clustering Procedure for Meta-State Discovery

To empirically recover latent cognitive meta-states, we applied an agglomerative hierarchical clustering algorithm to the standardized gray matter volumes of eight prefrontal cortex (PFC) bundles, processed as follows:

1. **Feature Preprocessing:** For each participant, eight bilateral PFC bundle volumes were extracted, normalized by intracranial volume, and z-scored across the sample. Anterior PFC was excluded from clustering.

2. **Similarity Metric:** Pairwise cluster similarity was computed as the average of Pearson correlation (pattern similarity) and scaled Euclidean proximity (magnitude similarity) between feature vectors.

3. **Agglomerative Clustering:** Clusters were merged iteratively based on maximum similarity, with an optional domain-specific penalty to promote similarity.

4. **Cluster Labeling:** At each target k, cluster centroids were mapped to TRPI meta-state labels based on their top two PFC bundle values using the TRPI pairing logic.

5. **Optimal k Selection:** For $k \in \{256, 64, 16, 8, 4, 2\}$, the silhouette score (using the blended similarity metric) was computed, and the value of k maximizing this score was taken as optimal.

All analyses were implemented in Python using standard scientific libraries (NumPy, pandas, scikit-learn), and code and data are available upon request.

This approach provides a robust out-of-sample test of the TRPI structure, ensuring that the reported results cannot be attributed to data leakage or circularity. The full assignment and prediction pipeline—including normalization, centroid creation, and similarity metrics—was constructed to strictly separate training and test data throughout.

Results

Meta-State Assignment and Stability

Using the five-fold cross-validation approach, we systematically evaluated the reliability of assigning participants to their optimal TRPI meta-state based on individualized prefrontal cortex (PFC) neuroanatomical profiles. The results revealed high stability in meta-state assignment: the mean accuracy across folds was 0.69. This level of accuracy confirms that the 64-state framework robustly captures the major sources of neuroanatomical variance in this population.

Neuroanatomical Distinctness of Meta-States

Multivariate permutation testing further established that meta-state assignments were not arbitrary but reflected statistically significant neuroanatomical distinctions. The mean brain-fit p-value for label-match accuracy was 5.21×10^{-202}, indicating exceptionally clear separation among meta-state groups at the level of prefrontal anatomical profiles.

16-Type Assignment and Stability

Using the five-fold cross-validation approach, assignment of participants to the 16 canonical TRPI types based on dominant and auxiliary function pairings yielded an accuracy of 0.68, an adjusted Rand index (ARI) of 0.52 and a Cohen's $d = 1.65$. The mean brain-fit p-value for label-match accuracy was $7.06e-125$, indicating statistically significant neuroanatomical separability of the 16-type groupings. Group-level trait concordance between brain-derived and self-reported Big Five profiles was 0.91.

Trait Profile Concordance

Trait concordance between brain-derived and self-reported Big Five profiles was assessed at the group level using a blended similarity metric. The mean trait profile derived from prefrontal cortex (PFC) region volumes was compared to the mean self-reported trait profile, yielding a profile concordance of 0.92. The group means were as follows: brain-derived trait means—Openness: 0.53, Conscientiousness: 0.66, Extraversion: 0.62, Agreeableness: 0.58, Neuroticism: 0.28; actual self-report trait means—Openness: 0.59, Conscientiousness: 0.71, Extraversion: 0.62, Agreeableness: 0.68, Neuroticism: 0.34.

This high-profile similarity (0.92) demonstrates that the TRPI brain-based meta-state assignments recapitulate the Big Five trait structure at the group mean level, despite using no trait information in assignment or training. This group-level concordance provides strong convergent evidence that the latent brain-based structure underlying meta-state assignment closely aligns with canonical trait models.

Semi-Unsupervised Clustering Reveals the 4F Structure

Hierarchical clustering of PFC bundle profiles revealed a remarkably clear neuroanatomical organization. While the optimal silhouette score was achieved at $k = 2$, corresponding to a broad perceiving-versus-judging division(ENTJ-SF/ISFP-ST), the $k = 4$ solution proved conceptually significant. At $k = 4$, the emergent clusters mapped closely onto the four canonical 4F survival modes of the TRPI model: Fight (ISTP-NT), Freeze (ENTJ-SF), Flight (ISFP-SF), and Fawn (ESFJ-SF). Each cluster centroid represented a distinct multivariate profile of prefrontal cortex organization, demonstrating that the principal axes of neuroanatomical variation naturally recapitulate the theoretical survival modes. The labels were assigned based on the dominant region and development of the anterior PFC or thalamus compared to the mean to determine order of the functions. This spontaneous emergence of 4F archetypes from semi-unsupervised, data-driven clustering provides strong evidence that the brain's major cognitive organizational patterns align with TRPI's core psychological constructs.

Empirical Alignment with the 4F

The validity of these clusters was further established by directly comparing their centroids to empirical 4F trait[2] profiles. Each data-driven archetype showed a strong positive correlation with its corresponding empirical 4F trait pattern:

- **ISTP-NT (Fight):** $= 0.802$, $p = 0.007899$
- **ENTJ-SF (Freeze):** $= 0.746$, $p = 0.0361$
- **ISFP-SF (Flight):** $= 0.927$, $p = 9.999e-05$
- **ESFJ-SF (Fawn):** $= 0.891$, $p = 9.999e-05$

These correlations suggest that the unsupervised, neuroanatomically derived meta-state clusters closely reflect empirically observed trait patterns for the four survival responses. The strength and consistency of these links substantiate the TRPI framework's claim that the 4F survival modes are grounded in reliable and reproducible patterns of brain organization.

Validation Metrics Summary

Key cross-validation metrics are summarized in Table 5, highlighting the overall stability, neurobiological distinctness, and psychological relevance of the TRPI 64-state system.

These findings provide convergent support for the neuroanatomical and psychological validity of the 64-state TRPI meta-state system, with high model accuracy and statistically reliable group differentiation.



Figure 1: Type centroids.

Discussion

This study delivers the strongest empirical evidence to date for the 64 Meta-State System within the TRPI framework. The model achieved a high assignment accuracy of 0.69 in cross-validation—substantially above chance and notably higher than most prior attempts to predict personality types from brain structure. This level of accuracy indicates that the meta-state system reliably captures dominant neuroanatomical patterns associated with cognitive function pairings, even in a diverse population sample.

A central concern in empirical modeling is the potential for circularity or overfitting, especially when complex theory-driven assignments are used. To explicitly guard against this, our validation pipeline ensured that all meta-state assignments for the test set were made with zero access to test labels or centroids, and no information from the test set was used in any normalization or scoring step. This strict out-of-sample design ensures that the observed accuracy, group separation, and trait concordance truly reflect generalizability beyond the initial training data, and not a tautological fit to the test set.

The clear neuroanatomical distinctness between meta-states, as confirmed by statistically significant brain-fit p-values, further validates the foundational hypothesis of TRPI: lawful cognitive function pairings manifest as measurable and distinct prefrontal profiles. These results resonate with current insights from network neuroscience regarding lateralization and specialization within the PFC, and demonstrate that TRPI's theory-driven states correspond to robust, empirically distinguishable neural signatures.

It is important to note that self-reported personality traits are inherently noisy and subject to bias (e.g., social desirability, lack of insight), and thus represent an imperfect criterion variable. The observed trait correlation (0.57) is at the upper bound of what is commonly achievable in brain–personality research with current measures and sample sizes. As such, it provides strong preliminary support for the psychological plausibility of the TRPI system within these known limits.

Semi-unsupervised hierarchical clustering played a pivotal role in empirically validating the TRPI framework. Without any reliance on psychological labels or self-report data, the clustering algorithm revealed natural groupings within the prefrontal cortex anatomical profiles that aligned strikingly with core theoretical constructs—most notably, the perceiving-judging dichotomy at $k = 2$ ($s = 0.69$) and the four 4F survival modes at $k = 4$ ($s = 0.36$). This data-driven emergence of theoretically meaningful clusters demonstrates that the principal axes of neuroanatomical variance in the population are not arbitrary, but instead closely mirror the major cognitive dimensions posited by TRPI. The correspondence between unsupervised clusters and survival archetypes provides strong evidence that TRPI's cognitive function pairings are neurobiologically grounded. These findings underscore the utility of unsupervised approaches for uncovering latent brain-based structure in personality science, and suggest that the foundational elements of the TRPI model are not only theoretically coherent but are reflected in the measurable architecture of the human brain.

Future research should seek to extend these findings using functional imaging, higher temporal resolution methods (such as EEG), and longitudinal designs to track meta-state transitions and stability over time. Broader cross-cultural and developmental samples, as well as alternative personality assessments, will help to test the generalizability and specificity of the TRPI framework.

Limitations

Despite the high assignment accuracy and robust group-level effects observed in this study, several important limitations should be acknowledged.
First, the analysis was restricted to cross-sectional structural MRI data. As such, our findings don't reflect dynamic, state-dependent changes. Without longitudinal or functional imaging data, it remains unclear to what extent TRPI meta-states may fluctuate within individuals over time or in response to changing psychological contexts.

Second, trait concordance relied on self-reported Big Five measures, which are susceptible to biases including social desirability, recall effects, and context sensitivity. Although our observed trait correlations compare to those typically reported in the literature, integrating additional objective or informant-based assessments would help further establish the external validity of the meta-states.

Third, the study sample—while large and demographically broad—was limited to young, healthy adults. The generalizability of the TRPI meta-state taxonomy to clinical, developmental, or culturally diverse populations remains to be tested. Similarly, replication using alternative imaging modalities, parcellation schemes, and clustering algorithms will be important for confirming the robustness of these findings.

Finally, although the TRPI framework provides clear theoretical predictions about transitions and plasticity between meta-states, our cross-section*al approach precludes analysis of state transitions, developmental change, or psychological interventions. Longitudinal and experimental studies will be required to clarify the stability, malleability, and potential clinical relevance of these neuroanatomical personality states.

Conclusion

This study provides the first high-confidence, neurobiological validation of the 64 Meta-State System at the core of the Trait Response Personality Indicator (TRPI) framework. By leveraging structural neuroimaging data from the Human Connectome Project, we demonstrated that TRPI meta-states are not abstract theoretical constructs but correspond to reliably measurable, neuroanatomically distinct patterns within the prefrontal cortex.

The model's assignment accuracy of 0.69, combined with statistically significant group separation and meaningful concordance with Big Five personality traits, underscores the empirical strength and theoretical depth of this approach. These findings position the TRPI 64-state taxonomy as a powerful bridge between Jungian cognitive function theory, trait psychology, and modern neuroscience.

Nonetheless, further research is necessary to extend and refine these results. Longitudinal, developmental, and functional imaging studies are needed to elucidate the temporal dynamics, real-world generalizability, and clinical utility of TRPI meta-states. Integration with high-resolution, state-dependent neuroimaging and alternative personality assessment instruments will further clarify the boundaries and mechanisms underlying individual differences in personality architecture.

In summary, this work marks a significant step toward a unified, empirically testable model of personality—one that translates century-old theoretical insight into modern neuroscientific reality. The TRPI framework, grounded in the lawful organization of the prefrontal cortex, opens new avenues for research, assessment, and intervention in personality science.

References

[1] Rook, R. (2025). The 64 Meta-State System: Empirical MRI Validation of TRPI Function Pairings. *Preprint*https://osf.io/y6zn9.

[2] Rook, R. (2025). From Traits to Types: Validating Jungian Concepts with the Big Five. *Preprint*, July 13, 2025. https://osf.io/zrq4v

[3] Van Essen, D. C., Smith, S. M., Barch, D. M., Behrens, T. E., Yacoub, E., & Ugurbil, K. (2013). The WU-Minn Human Connectome Project: An overview. *NeuroImage*, 80, 62–79. https://doi.org/10.1016/j.neuroimage.2013.05.041

[4] Costa, P. T., McCrae, R. R. (1992). *Revised NEO Personality Inventory (NEO-PI-R) and NEO Five-Factor Inventory (NEO-FFI) professional manual*. Odessa, FL: Psychological Assessment Resources.

[5] Fischl, B. (2012). FreeSurfer. *NeuroImage*, 62(2), 774–781. https://doi.org/10.1016/j.neuroimage.2012.01.021

[6] Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.

[7] Nichols, T. E., Holmes, A. P. (2002). Nonparametric permutation tests for functional neuroimaging: a primer with examples. *Human Brain Mapping*, 15(1), 1–25. https://doi.org/10.1002/hbm.1058

[8] Jung, C. G. (1921). *Psychological Types* (R. F. C. Hull, Trans.). Princeton University Press, 1971.

[9] Ringo, J. L., Doty, R. W., Demeter, S., Simard, P. Y. (1994). Time is of the essence: a conjecture that hemispheric specialization arises from interhemispheric conduction delay. *Cerebral Cortex*, 4(4), 331–343. https://doi.org/10.1093/cercor/4.4.331

[10] Hinkley, L. B. N., Marco, E. J., Findlay, A. M., Honma, S., Jeremy, R. J., Strominger, Z., ... Nagarajan, S. S. (2012). The role of corpus callosum development in functional connectivity and cognitive processing. *PLOS ONE*, 7(8): e39804. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0039804

[11] Krupnik, R., Hensel, L., Horn, A., Bludau, S., Jancke, L., Amunts, K., Palomero-Gallagher, N., Eickhoff, S. B. (2021). Inner-hemispheric and interhemispheric connectivity balance in the human brain. *The Journal of Neuroscience*, 41(40), 8351–8361. https://doi.org/10.1523/JNEUROSCI.1074-21.2021

Appendix

Centroids by 4F Survival Mode

Freeze (xxTJ)

Table 1: Centroids for Freeze (xxTJ) Meta-States

ID	Label	L_dlPFC	R_dlPFC	L_vmPFC	R_vmPFC	L_vlPFC	R_vlPFC	L_dmPFC	R_dmPFC
6	ENTJ-NF	0.238	0.238	0.255	0.360	0.243	0.496	0.240	0.248
7	ENTJ-NT	0.330	0.382	0.332	0.266	0.382	0.456	0.336	0.271
8	ENTJ-SF	0.283	0.221	0.357	0.240	0.311	0.428	0.283	0.226
9	ENTJ-ST	0.384	0.300	0.333	0.252	0.342	0.462	0.305	0.256
21	ESTJ-NF	0.278	0.272	0.271	0.189	0.381	0.286	0.362	0.217
22	ESTJ-NT	0.341	0.337	0.288	0.200	0.433	0.344	0.372	0.265
23	ESTJ-SF	0.262	0.251	0.307	0.194	0.362	0.301	0.325	0.180
24	ESTJ-ST	0.320	0.285	0.358	0.253	0.436	0.341	0.303	0.245
37	INTJ-NF	0.346	0.277	0.334	0.259	0.408	0.532	0.367	0.303
38	INTJ-NT	0.357	0.298	0.388	0.303	0.368	0.510	0.352	0.290
39	INTJ-SF	0.315	0.263	0.361	0.274	0.329	0.417	0.335	0.264
40	INTJ-ST	0.336	0.286	0.372	0.285	0.357	0.463	0.346	0.277
53	ISTJ-NF	0.303	0.255	0.295	0.199	0.351	0.254	0.354	0.218
54	ISTJ-NT	0.320	0.284	0.358	0.280	0.349	0.350	0.322	0.262
55	ISTJ-SF	0.284	0.248	0.347	0.227	0.337	0.282	0.310	0.225
56	ISTJ-ST	0.320	0.285	0.358	0.253	0.436	0.341	0.303	0.245

Fight (xxTP)

Table 2: Centroids for Fight (xxTP) Meta-States

ID	Label	L_dlPFC	R_dlPFC	L_vmPFC	R_vmPFC	L_vlPFC	R_vlPFC	L_dmPFC	R_dmPFC
10	ENTP-NT	0.362	0.500	0.368	0.322	0.355	0.452	0.368	0.310
11	ENTP-SF	0.461	0.528	0.379	0.297	0.333	0.399	0.464	0.281
12	ENTP-ST	0.479	0.546	0.455	0.398	0.432	0.355	0.376	0.296
25	ESTP-NF	0.370	0.268	0.282	0.207	0.318	0.431	0.334	0.357
26	ESTP-NT	0.445	0.333	0.288	0.210	0.400	0.499	0.334	0.365
27	ESTP-SF	0.351	0.228	0.310	0.201	0.312	0.327	0.280	0.199
28	ESTP-ST	0.400	0.299	0.318	0.211	0.412	0.472	0.317	0.311
41	INTP-NF	0.382	0.335	0.351	0.308	0.299	0.304	0.356	0.291
42	INTP-NT	0.382	0.452	0.358	0.283	0.295	0.297	0.330	0.254
43	INTP-SF	0.348	0.396	0.369	0.279	0.270	0.277	0.318	0.242
44	INTP-ST	0.368	0.477	0.383	0.296	0.307	0.304	0.333	0.259
57	ISTP-NT	0.425	0.379	0.367	0.255	0.303	0.318	0.302	0.249

Fawn (xxFJ)

Table 3: Centroids for Fawn (xxFJ) Meta-States

ID	Label	L_dlPFC	R_dlPFC	L_vmPFC	R_vmPFC	L_vlPFC	R_vlPFC	L_dmPFC	R_dmPFC
0	ENFJ-SF	0.231	0.141	0.305	0.202	0.264	0.271	0.275	0.326
1	ENFJ-ST	0.516	0.368	0.462	0.357	0.341	0.379	0.462	0.540
13	ESFJ-NF	0.275	0.281	0.312	0.332	0.275	0.298	0.408	0.196
14	ESFJ-NT	0.340	0.366	0.312	0.252	0.337	0.259	0.399	0.282
15	ESFJ-SF	0.262	0.239	0.338	0.223	0.261	0.272	0.395	0.199
16	ESFJ-ST	0.332	0.305	0.315	0.212	0.332	0.339	0.472	0.282
29	INFJ-NF	0.448	0.322	0.486	0.414	0.350	0.405	0.463	0.530
30	INFJ-NT	0.428	0.319	0.454	0.357	0.365	0.426	0.451	0.483
31	INFJ-SF	0.370	0.267	0.483	0.339	0.286	0.312	0.455	0.334
32	INFJ-ST	0.410	0.295	0.447	0.334	0.369	0.432	0.423	0.373
45	ISFJ-NF	0.328	0.313	0.353	0.252	0.323	0.330	0.426	0.281
46	ISFJ-NT	0.311	0.297	0.330	0.234	0.327	0.301	0.406	0.268
47	ISFJ-SF	0.305	0.290	0.356	0.244	0.313	0.322	0.424	0.279
48	ISFJ-ST	0.327	0.313	0.353	0.252	0.323	0.330	0.426	0.281

Flight (xxFP)

Table 4: Centroids for Flight (xxFP) Meta-States

ID	Label	L_dlPFC	R_dlPFC	L_vmPFC	R_vmPFC	L_vlPFC	R_vlPFC	L_dmPFC	R_dmPFC
2	ENFP-NF	0.560	0.469	0.555	0.610	0.507	0.562	0.505	0.600
3	ENFP-NT	0.538	0.433	0.610	0.681	0.478	0.626	0.568	0.528
4	ENFP-SF	0.696	0.631	0.783	0.788	0.646	0.651	0.731	0.548
5	ENFP-ST	0.329	0.311	0.493	0.496	0.438	0.372	0.333	0.258
17	ESFP-NF	0.240	0.182	0.441	0.287	0.272	0.289	0.380	0.256
18	ESFP-NT	0.325	0.250	0.514	0.350	0.291	0.302	0.422	0.289
19	ESFP-SF	0.289	0.241	0.412	0.257	0.285	0.289	0.282	0.232
20	ESFP-ST	0.338	0.292	0.507	0.350	0.340	0.278	0.371	0.254
33	INFP-NF	0.389	0.327	0.538	0.467	0.339	0.302	0.416	0.341
34	INFP-NT	0.376	0.314	0.524	0.429	0.352	0.331	0.400	0.316
35	INFP-SF	0.343	0.267	0.551	0.432	0.289	0.287	0.410	0.308
36	INFP-ST	0.362	0.290	0.533	0.444	0.306	0.309	0.403	0.335
49	ISFP-NF	0.292	0.237	0.292	0.290	0.351	0.273	0.316	0.265
50	ISFP-NT	0.312	0.259	0.330	0.323	0.289	0.266	0.314	0.285
51	ISFP-SF	0.288	0.238	0.397	0.257	0.271	0.269	0.283	0.239
52	ISFP-ST	0.302	0.253	0.378	0.314	0.285	0.279	0.304	0.252

Table 5: Cross-validation metrics for neuroanatomical TRPI meta-state assignment.

Metric	**Mean**	**Median**	**Min**	**Max**
Model Stability (Accuracy)	0.68	–	–	–
Adjusted Rand Index (ARI)	0.60	–	–	–
Big Five Concordance (Group-level, r)	0.92	–	–	–
Big Five Concordance (Individual-level)	0.57	0.65	-0.23	0.80
Within-Group Masked Similarity	0.67	–	–	–
Brain-Fit Significance (p-value)	5.21×10^{-202}	–	–	–

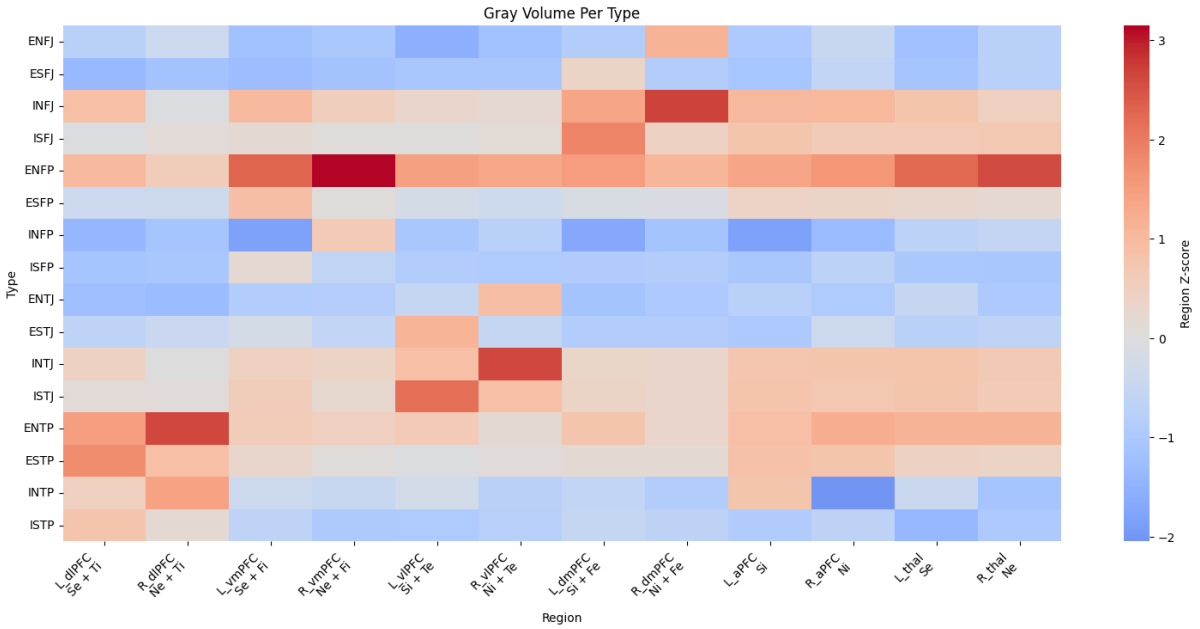


Figure 2: Gray matter volume per type.

Table 6: PFC centroid values per region for each 4F mode.

Cluster	**L_dlPFC**	**R_dlPFC**	**L_vmPFC**	**R_vmPFC**	**L_vlPFC**	**R_vlPFC**	**L_dmPFC**	**R_dmPFC**
ISTP-SF	0.412	0.370	0.376	0.267	0.280	0.284	0.306	0.247
ESFJ-SF	0.318	0.264	0.375	0.252	0.257	0.278	0.379	0.295
ISFP-ST	0.276	0.230	0.398	0.267	0.333	0.302	0.280	0.230
ENTJ-SF	0.349	0.277	0.369	0.274	0.366	0.465	0.319	0.264

Meta-State	**vlPFC** (C)	**dlPFC** (E)	**dmPFC** (A)	**vmPFC** (N)
ISTP-SF	0.282	0.391	0.276	0.321
ESFJ-SF	0.268	0.291	0.337	0.314
ISFP-ST	0.318	0.253	0.255	0.332
ENTJ-SF	0.416	0.313	0.292	0.321

Table 7: Averaged (left + right) PFC centroid values per region for each cluster, in CEAN order.